Deterministic Connectivity: Audited Baseline and Collision Barriers

Research addendum, 29 October 2026

Abstract

We give a self-contained proof of the supplied draft's deterministic $\Omega(\sqrt{n})$ lower bound for exact Boolean connectivity in the simultaneous vertex-neighborhood model, with explicit alphabet and bit constants. We also establish limits on several proposed collision arguments: local linear maps with $r + \lceil \log_2 n \rceil$ bits exclude every nonzero adjacency difference of binary rank at most r; local XOR-linear cut-space recognizers have concatenated rank at least $\binom{m}{2} - 1$; a four-vertex example defeats a particular anchored collision template; and public r-sheet lifts have a direct reconstruction protocol. These results do not improve the unrestricted $\Omega(\sqrt{n})$ lower bound. This is an independent addendum, with no claim of novelty or author acceptance.

1 Definitions and scope

An input is a simple undirected graph G on the public labeled vertex set $[n]$. Player v knows exactly its own neighborhood $N_G(v)$ and sends one deterministic message to a referee. All messages are simultaneous. Encoders may be vertex-specific and nonlinear, and the referee has no computational restriction. The required output is the single bit indicating whether G is connected, with correctness on every input. No shared knowledge of other vertices' neighborhoods is available.

The communication cost B is the worst-case *maximum bits sent by one vertex*, not the total number of bits. We distinguish a message alphabet of cardinality at most 2^b, fixed-length binary messages, and arbitrary binary strings of maximum length B. All logarithms are base two unless written ln. Only ordinary connectivity, or $k = 1$, is treated here.

For algebraic statements, edge indicators lie in $\mathbb{F}_2^{\binom{n}{2}}$, with the fixed ambient edge set $\binom{[n]}{2}$. An adjacency difference D is a symmetric zero-diagonal matrix over $\mathbb{F}_2$; D_v denotes its incident row with the diagonal coordinate omitted. The ranks of these adjacency matrices and the ranks of concatenated message maps are different quantities.

2 The unrestricted connectivity baseline

Theorem 2.1 (D-01, ordinary connectivity)**.** *Let $n \geq 2$ and let $b \geq 1$ be an integer. If an exact deterministic connectivity protocol has at most 2^b possible messages at each vertex, then*

$$b > \frac{n}{768(\lceil \sqrt{n} \rceil + 1)} > \frac{\sqrt{n}}{2304}. \tag{1}$$

Consequently, fixed-length messages of maximum length B satisfy $B > \sqrt{n}/2304$. Variable-length binary strings of maximum length B, including the empty string, satisfy $B > \sqrt{n}/4608$.

We first prove the counting fact used to select a common partition. A family $\mathcal{S} \subseteq \{1,2,3,4\}^q$ *shatters* a coordinate set if its projection onto those coordinates contains every assignment.

Lemma 2.2 (Four-symbol shattering bound)**.** *If $\mathcal{S} \subseteq \{1,2,3,4\}^q$ shatters no coordinate set of size d, then*

$$|\mathcal{S}| \le F(q,d) := \sum_{j=0}^{d-1} \binom{q}{j} 3^{q-j}.$$

Here $F(q,0) = 0$ and $F(q,d) = 4^q$ for $d > q$.

Proof. The cases $d = 0$ and $d > q$ are immediate, since any nonempty family shatters the empty coordinate set. Induct on q, with $q = 0$ immediate. Let $\mathcal{S}_a$ be the slice having last coordinate a, projected onto the first $q-1$ coordinates. Set $\mathcal{U} = \bigcup_{a=1}^{4} \mathcal{S}_a$ and $\mathcal{I} = \bigcap_{a=1}^{4} \mathcal{S}_a$. Each projected word occurs in at most three slices unless it is in $\mathcal{I}$, so

$$|\mathcal{S}| = \sum_{a=1}^{4} |\mathcal{S}_a| \le 3|\mathcal{U}| + |\mathcal{I}|.$$

The family $\mathcal{U}$ cannot shatter d coordinates. The family $\mathcal{I}$ cannot shatter $d-1$ coordinates, since adjoining the last coordinate would then give a shattered d-set in $\mathcal{S}$. Induction and Pascal's identity give

$$|\mathcal{S}| \le 3F(q-1,d) + F(q-1,d-1) = F(q,d).$$

□

Proof of Theorem 2.1. Fix a correct protocol and put

$$q = 128(\lceil \sqrt{n} \rceil + 1), \qquad M = n - q.$$

If $b > \sqrt{n}$, the asserted alphabet bound is immediate, so assume $b \le \sqrt{n}$. If $M < 4(qb+1)$, then, using $q \ge 4$ and $b \ge 1$,

$$n < (4b+1)q + 4 \le (4b+2)q \le 6bq, \tag{2}$$

as desired. We therefore suppose, towards a contradiction, that

$$M \ge 4(qb+1). \tag{3}$$

This implies $q < n$. Choose a reservoir U of q vertices and designate the remaining M vertices as centers $c_1, \dots, c_M$. For each center let $f_i(S)$ be its actual message when its neighborhood is $S \subseteq U$.

Local witnesses. Set $d = 16(b+1)$. Consider any d-element set $Y \subseteq U$ and one center c_i. Pair the vertices of Y and designate a pair $\{\alpha, \beta\}$. Consider all sets A containing α, excluding β, and choosing exactly one vertex of each other pair. There are $2^{d/2-1} = 2^{8b+7} > 2^{2b}$ such sets. The ordered message pair $(f_i(A), f_i(Y \setminus A))$ has at most 2^{2b} values. Thus distinct sets A_0, A_1 have the same ordered pair of messages. Define

$$\begin{aligned} P_i^0 &= A_0 \cap A_1, \quad P_i^1 = Y \setminus (A_0 \cup A_1), \\ R_i^0 &= A_0 \setminus A_1, \quad R_i^1 = A_1 \setminus A_0. \end{aligned}$$

These sets partition Y. The first two contain α and β, respectively. The last two are nonempty because the two choices differ on at least one entire pair. Equality of the two ordered messages yields

$$\begin{aligned} f_i(P_i^0 \cup R_i^0) &= f_i(P_i^0 \cup R_i^1), \\ f_i(P_i^1 \cup R_i^0) &= f_i(P_i^1 \cup R_i^1). \end{aligned} \tag{4}$$

Label these four sets by P_0, P_1, R_0, R_1, respectively. Every extension of this labeling to U contains the same nonempty witnessing subsets.

One partition for every center. Call a labeling $U \to \{P_0, P_1, R_0, R_1\}$ good for c_i if it admits four nonempty subsets in the indicated classes satisfying (4), and bad otherwise. The local argument shows that the bad family shatters no d-element coordinate set. Under an independently uniform labeling of U, Lemma 2.2 gives

$$\Pr[\text{bad for } c_i] \le 4^{-q} \sum_{j<d} \binom{q}{j} 3^{q-j} = \Pr[Z < d], \qquad Z \sim \mathrm{Bin}(q, 1/4).$$

Since $b \le \sqrt{n}$, we have $d \le q/8$. Write $\mu = q/4$ and $t = \ln 2$. Exponential Markov's inequality implies

$$\begin{aligned} \Pr[Z \le \mu/2] &\le e^{t\mu/2} \mathbb{E} e^{-tZ} = e^{t\mu/2} (3/4 + e^{-t}/4)^q \\ &\le \exp\{\mu(t/2 - 1/2)\} \le e^{-\mu/8} = e^{-q/32}. \end{aligned}$$

Here $1 + x \le e^x$ and $(\ln 2)/2 - 1/2 < -1/8$ justify the last two steps. A union bound over the $M \le n$ centers gives

$$\Pr[\text{some center is bad}] \le n e^{-q/32} \le n e^{-4\sqrt{n}} < 1.$$

For the final inequality, setting $x = \sqrt{n} \ge 1$ gives $2 \ln x - 4x < 0$. Therefore a partition $U = P_0 \sqcup P_1 \sqcup R_0 \sqcup R_1$ is good for every center. Fix it and fix witnessing subsets for every center. No independence of the centers' bad events was used.

A global transcript collision. For $a, z \in \{0,1\}^M$, define $G(a, z)$ by making $P_0 \cup R_0$ and $P_1 \cup R_1$ into cliques and giving c_i exactly the neighborhood $P_i^{a_i} \cup R_i^{z_i}$. There are no other edges. Both cliques are nonempty. The graph $G(x, x)$ has exactly two components, because every center attaches within just one clique. If $x \ne y$, a center at a differing coordinate joins the two cliques in $G(x, y)$; all other centers have a neighbor in a clique. Thus $G(x, y)$ is connected.

By (4), each center's message depends only on a_i. A vertex in $P_0 \cup P_1$ sees its fixed clique and centers selected only by a, so its message depends only on a. A vertex in $R_0 \cup R_1$ analogously has a message depending only on z. Their joint transcript $T(z)$ has at most 2^{qb} values. Since (3) implies $M > qb$, there are distinct x, y with $T(x) = T(y)$. The full transcripts of $G(x, x)$ and $G(x, y)$ agree: the center and P messages agree because $a = x$ is fixed, and the R messages agree by construction. Their connectivity differs, a contradiction.

It follows that $M < 4(qb + 1)$, so (2) proves $b > n/(6q)$. This argument also covers all small n: when $q \ge n$, the inequality $M < 4(qb + 1)$ already holds and no reservoir is constructed. Finally $\lceil \sqrt{n} \rceil + 1 < 3\sqrt{n}$ for $n \ge 2$, which proves the second inequality in (1).

Zero-bit protocols are impossible, since the empty and complete graphs require different outputs. Fixed-length messages therefore have $B \ge 1$, and we apply the alphabet result with $b = B$. For variable-length strings, the available alphabet has size $\sum_{j=0}^{B} 2^j = 2^{B+1} - 1 < 2^{B+1}$. Apply the result with $b = B + 1$ and use $B + 1 \le 2B$ to obtain $B > \sqrt{n}/4608$. □

3 A barrier to bounded-rank collision witnesses

Theorem 3.1. *For integers $n \ge 2$ and $r \ge 0$, there are vertex-specific linear maps*

$$A_v : \mathbb{F}_2^{n-1} \longrightarrow \mathbb{F}_2^b, \qquad b = r + \lceil \log_2 n \rceil,$$

such that every nonzero symmetric zero-diagonal $n \times n$ matrix D of rank at most r satisfies $A_v(D_v) \neq 0$ for some vertex v. Consequently, two graph inputs with a nonzero adjacency-XOR difference of rank at most r have distinct concatenated transcripts under these maps.

The maps in this theorem are deterministic after being fixed by an existence argument. The theorem supplies neither a connectivity-deciding referee nor an efficient uniform construction of the maps. A chosen dense map can be evaluated in $O(nb)$ binary operations per vertex.

Lemma 3.2. *For $t \geq 1$ and $r \geq 0$, there are at most 2^{rt} symmetric zero-diagonal $t \times t$ matrices over $\mathbb{F}_2$ of rank at most r.*

Proof. Such a matrix represents an alternating bilinear form. If the form is nonzero, choose two vectors with pairing one. Their two-dimensional span is nondegenerate and splits off from its orthogonal complement. Iterating gives a basis in which the form is a direct sum of blocks $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and a zero block. Thus the rank is even, say $2k$, and the matrix can be written $UJ_{2k}U^{\mathsf{T}}$, where J_{2k} is the sum of k such blocks and U has t rows and $2k$ columns. This follows by transforming the displayed normal form back to the original basis.

Put $r' = 2\lfloor r/2 \rfloor$. Padding U with zero columns represents every matrix of rank at most r as $UJ_{r'}U^{\mathsf{T}}$ with $U \in \mathbb{F}_2^{t \times r'}$. There are $2^{r't} \leq 2^{rt}$ choices of U. The representation need not be unique. If $r' = 0$, it represents only the zero matrix, as required. □

Proof of Theorem 3.1. Choose every entry of every local matrix A_v independently and uniformly in $\mathbb{F}_2$. Fix a nonzero adjacency difference D and let T be its set of nonzero rows, with $t = |T|$. Each row D_v, $v \in T$, is nonzero, so its image under A_v is uniformly distributed in $\mathbb{F}_2^b$. The local maps are independent. Hence

$$\Pr[A_v(D_v) = 0 \text{ for all } v] = 2^{-bt}.$$

Symmetry confines every nonzero entry to the principal submatrix on T. Moreover $t \geq 2$, because a nonzero off-diagonal entry makes both its rows nonzero. Lemma 3.2 bounds the number of possible matrices on each fixed T by 2^{rt}. Union-bounding over actual row supports gives

$$\begin{aligned} \Pr[\text{some nonzero } D \text{ of rank } \leq r \text{ is killed}] &\leq \sum_{t=2}^{n} \binom{n}{t} 2^{(r-b)t} \\ &\leq \sum_{t=2}^{n} \binom{n}{t} n^{-t} \leq \sum_{t=2}^{n} \frac{1}{t!} < 1. \end{aligned}$$

For the strict inequality, $t! \geq 2^{t-1}$ gives the upper bound $\sum_{t=2}^{n} 2^{-(t-1)} = 1 - 2^{-(n-1)} < 1$. The count includes matrices whose support is smaller than T, which only enlarges the bound; the probability 2^{-bt} is assigned only to matrices with actual support T. Thus a successful choice of maps exists. For $r = 0, 1$ the claim is vacuous, since a nonzero alternating matrix has rank at least two. □

Remark 3.3. In particular, rank-two differences can all be excluded by $\lceil \log_2 n \rceil + 2$ bits per vertex. A universal collision claim using only the message budget therefore cannot require such differences for every encoder collection. This does not refute an argument using additional consequences of an assumed correct connectivity protocol, exclude high-rank collisions, or give an unrestricted connectivity upper bound. If $b > n - 1$, full-row transmission already provides a simpler separator.

4 Two cut-space facts with restricted conclusions

On the complete graph K_m, define the binary cut space

$$\mathcal{C}_m = \{\delta s : s \in \mathbb{F}_2^m\}, \qquad (\delta s)_{uv} = s_u + s_v.$$

In the local recognition problem, vertex v knows only the edge-label vector x_v on its star and sends a message; the referee must decide whether $x \in \mathcal{C}_m$. This is a distinct task from unrestricted graph connectivity.

Theorem 4.1 (Local XOR-linear cut recognition)**.** *Let $m \geq 3$. Suppose local linear maps $L_v : \mathbb{F}_2^{m-1} \to \mathbb{F}_2^{B_v}$, concatenated as L, permit an arbitrary referee to recognize $\mathcal{C}_m$ exactly. Then*

$$\operatorname{rank} L \geq \binom{m}{2} - 1.$$

For the maximum fixed-length message budget $B = \max_v B_v$, this gives

$$\sum_v B_v \geq \binom{m}{2} - 1, \qquad B \geq \left\lceil \frac{\binom{m}{2} - 1}{m} \right\rceil.$$

Proof. Let $K = \ker L$. Every vector in K has the same transcript as the zero cut, so correctness implies $K \subseteq \mathcal{C}_m$. Suppose $\dim K \geq 2$, and choose independent cuts $\delta s, \delta t \in K$. Define

$$x_{uv} = s_v t_u + t_v s_u \qquad (u \neq v).$$

This is a symmetric edge labeling. At vertex v,

$$x_v = s_v(\delta t)_v + t_v(\delta s)_v,$$

since the two copies of $s_v t_v$ cancel in characteristic two. Local linearity gives $L_v(x_v) = 0$ at every vertex, hence $x \in K$.

The labels $z_v = (s_v, t_v) \in \mathbb{F}_2^2$ affinely span $\mathbb{F}_2^2$. Otherwise some nonzero (a, b) and constant c satisfy $as_v + bt_v = c$ for every v, implying $a\delta s + b\delta t = 0$, contrary to independence. Choose p, q, r such that $z_q + z_p$ and $z_r + z_p$ are independent. Direct expansion then gives

$$x_{pq} + x_{qr} + x_{rp} = \det(z_q + z_p, z_r + z_p) = 1.$$

Every cut has triangle sum zero, so $x \notin \mathcal{C}_m$, contradicting $x \in K \subseteq \mathcal{C}_m$. Thus $\dim K \leq 1$, and rank-nullity gives the rank bound. Finally $\operatorname{rank} L \leq \sum_v B_v \leq mB$ proves the communication consequences. □

The decisive step uses locality and linearity of each encoder. The theorem imposes no linearity assumption on the referee, but its proof does not extend to arbitrary nonlinear encoders. Its primary conclusion concerns the total rank of L, rather than a rank bound at each vertex.

Proposition 4.2 (A four-vertex obstruction to an anchored template)**.** *Let $S = \{s \in \mathbb{F}_2^4 : s_0 = 0\}$ and define*

$$h_0(s) = s_1, \qquad h_1(s) = s_2, \qquad h_2(s) = s_3, \qquad h_3(s) = s_1.$$

There is no $s \in S$ and nonzero symmetric zero-diagonal matrix D with root row $D_0 = 0$ for which

$$h_v(s + D_v) = h_v(s) \quad \text{for every } v. \tag{5}$$

Here D_v is the full four-coordinate row, including its zero diagonal entry. Nevertheless, the associated local messages $h_v(P_v(x))$, where

$$P_v(x)_i = x_{vi} + x_{v0} \quad (0 \le i \le 3), \qquad x_{vv} = 0,$$

do not recognize $\mathcal{C}_4$.

Proof. The anchor condition makes $D_{01} = D_{02} = D_{03} = 0$, and also ensures that every $s + D_v$ belongs to S. Since the h_v are linear, (5) at vertices $1, 2, 3$ forces, respectively, $D_{12} = D_{23} = D_{31} = 0$. These are all six possible edges, so $D = 0$.

The map P_v depends only on the local star and always has coordinate zero equal to zero. On a cut $x = \delta s$ with $s_0 = 0$, it satisfies $P_v(x) = s$. For an arbitrary x, the four messages are

$$(x_{01},\ x_{12} + x_{10},\ x_{23} + x_{20},\ x_{31} + x_{30}).$$

Set $x_{02} = x_{23} = 1$ and all other edge labels to zero. The transcript is zero, as for the zero cut. But the triangle $(0, 1, 2)$ has label sum one, so x is not a cut. No referee receiving these messages can recognize $\mathcal{C}_4$ exactly. □

This is an exact finite counterexample to the specified zero-root-row template. It does not refute an asymptotic nonlinear lower bound or show that every possible cut/noncut collision has been excluded.

5 The cost of describing public graph lifts

Proposition 5.1 (Public-base reconstruction)**.** *Fix a public simple base graph H on m labeled vertices and an integer $r \ge 2$. The input has public labels $(v, a) \in V(H) \times [r]$ and is promised to be an r-sheet lift: every base edge is replaced by a perfect matching between its two fibers, and there are no other edges. There is an exact deterministic simultaneous protocol reconstructing the entire input, and hence deciding connectivity, with at most*

$$\Delta(H)\,\lceil \log_2 r \rceil$$

bits per lifted vertex and polynomial decoding time.

Proof. Order the neighbors of each base vertex using the public labels. Vertex (v, a) has exactly one neighbor in each fiber u with $u \in N_H(v)$. In that order, it sends the sheet index of each such neighbor, using $\lceil \log_2 r \rceil$ bits per index. Its message is computed from its own neighborhood. The referee knows the order and the base labels, so it reconstructs all edges from these indices. Ordinary graph search decides connectivity. The encoding and decoding are polynomial in the explicit input and output sizes. □

For $H = K_m$ and $n = mr$, the budget is $(m-1)\lceil \log_2 r \rceil = O((n/r)\log r)$. Along a growing sequence of such families, for any fixed $C \ge 0$, the condition

$$\frac{r}{\log r} = \omega(\log^C n)$$

makes this budget $o(n/\log^C n)$ and therefore rules out an $\Omega(n/\log^C n)$ lower bound on that promise family. The public base and public fiber labels are essential assumptions. The proposition supplies no such protocol when these data are hidden, and no upper bound for arbitrary graphs.

Provenance and open scope

The baseline is the $k = 1$ specialization of the supplied original draft, `sources/draft/paper-body.pdf`, Section 6, pages 12–16. Its explicit reconstruction is in `results/agents/lower_bounds/2026-10-29-analysis.md`. The four-symbol bound is proved here in full; the draft attributes its general form to Karpovsky–Milman. No external citation is needed for the arguments in this note.

The separate proof audit is recorded in `results/agents/proof_auditor/2026-10-29-blind.md` and `results/agents/proof_auditor/2026-10-29-audit.md`. The blind report independently proved the cut-rank bound, the four-vertex counterexample, the public-lift protocol, and a version of Theorem 3.1 with two extra output bits. It initially left the central baseline unresolved; the subsequent audit checks the released self-contained baseline proof.

Theorem 3.1 uses the auditor's symplectic counting proof and its subsequent refinement removing those two bits. The refinement also received a separate reverse audit in `results/agents/algebraic_sketches/2026-10-29-rank-constant-audit.md`. The original algebraic analysis is `results/agents/algebraic_sketches/2026-10-29-analysis.md`. These records document independent checks within this research run; they do not establish publication priority or author acceptance.

An unrestricted improvement beyond $\Omega(\sqrt{n})$ remains open in this work. General-k claims, randomized protocols, stronger nonlinear collision conjectures, and unselected prior-note assertions receive no verdict here. This document is independently prepared; the original author's editable source was not supplied, and the source PDF is not modified.